Umar Rahi

MCA Graduate | Entry-Level Data Science and Machine Learning
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PROFESSIONAL SUMMARY

MCA graduate (2026) with a foundation in Python, machine learning, data preprocessing, and predictive modeling. Built a Random Forest AQI prediction service with a Flask REST API and a leakage-aware churn pipeline comparing Logistic Regression, Decision Tree, and Random Forest models. Completed IBM SkillsBuild Data Analytics with AI Internship Program and IBM modules on LLMs, RAG, and generative AI. Seeking entry-level Data Scientist or Machine Learning roles.

TECHNICAL SKILLS

Programming and Databases: Python, SQL (MySQL, Oracle SQL, SQLite)
Data Science: Pandas, NumPy, Exploratory Data Analysis, Data Preprocessing, Feature Engineering, Descriptive Statistics
Machine Learning: Supervised Learning (Regression, Classification), Random Forest, Logistic Regression, Decision Trees, Scikit-learn Pipelines, Model Training and Evaluation (train-test split, MSE, RMSE, R^2), Feature Importance
Generative AI: LLM API Integration (OpenRouter, OpenAI SDK); LLM and RAG fundamentals (IBM SkillsBuild coursework)
Development and Tools: Flask, REST APIs, Streamlit, Plotly, Matplotlib, Seaborn, Jupyter Notebook, Git, GitHub, pytest

PROJECTS

AQI Predictor: Air Quality Index Web Application GitHub | Live Demo
Python, Scikit-learn, Pandas, NumPy, Joblib, Flask, Chart.js

- Generated a 5,000-record synthetic dataset of eight pollutant and weather features, with AQI labels derived from simplified EPA-style breakpoints; applied StandardScaler and an 80/20 train-test split.
- Trained a Random Forest Regressor (100 trees, max depth 15), evaluated it with MSE, RMSE, and R^2, and exported feature importances; serialized the model and scaler with Joblib.
- Developed a Flask REST API with input validation, bounded predictions (0–500), and AQI category mapping, served through a Chart.js front end and deployed on Render.

Customer Churn Analysis and Prediction (IBM SkillsBuild Internship Project) GitHub
Python, Pandas, Scikit-learn, Plotly, Streamlit, pytest

- Cleaned and merged three relational worksheets (21 customers) and engineered features including tenure, age group, and support-ticket aggregates; defined the churn target from cancellation dates.
- Built a scikit-learn preprocessing pipeline (imputation, scaling, one-hot encoding) and trained class-weighted Logistic Regression, Decision Tree, and Random Forest models; excluded leakage-prone fields such as cancellation reason and churn score.
- Labeled all metrics as in-sample and documented that 21 records are too few for holdout evaluation; presented risk scores and feature importance in a Streamlit dashboard.

NexusAI: LLM Chatbot Application GitHub | Live Demo
Python, Flask, OpenRouter API, OpenAI SDK, SQLite, Gunicorn

- Built a full-stack chatbot using the OpenRouter API through the OpenAI Python SDK, with Server-Sent Events streaming, multi-turn context, and Markdown rendering.
- Implemented SQLite-backed conversation history (create, rename, delete, search), per-client rate limiting, server-side API key handling, and pytest tests; deployed on Render.

EDUCATION

Master of Computer Applications (MCA), Graphic Era University, Dehradun | 2024 – 2026
Bachelor of Computer Applications (BCA), Graphic Era Hill University, Dehradun | 2021 – 2024

CERTIFICATIONS

- IBM SkillsBuild Data Analytics with AI Internship Program (6 weeks), AICTE and BharatCares, Aug – Sep 2026
- Generative AI Essentials: Using LLMs to Work with Data, IBM SkillsBuild digital badge
- Data Fundamentals; Getting Started with Data, IBM SkillsBuild digital badges
- IBM SkillsBuild learning modules: Introduction to Large Language Models; Introduction to Retrieval-Augmented Generation; Classifying Data Using IBM Granite